Junsheng Huang

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github.com/no-touch-fish

Education

ZheJiang University-University of Illinois at Urbana-Champaign Aug. 2021 - Jun. 2025

Dual Bachelor of Electrical and Computer Engineering (UIUC GPA: 4.00 / 4.00) (ZJU GPA: 3.97 / 4.00)

- **Relevant Courses:** Computer System Engineering(ECE391), Data Structure(CS225), Artificial Intelligence(CS440), Algorithm(CS374), Machine Learning(CS446), Applied Parallel Programming(ECE408)

University of Illinois at Urbana-Champaign Aug. 2025 – May 2027 (expected)

Master of Science in Electrical and Computer Engineering

- **Relevant Courses:** Reinforce Learning (ECE586), Natural Language Processing (CS546), Computer Vision (CS543), Graph Machine Learning (CS598)

Research Experience

SAFTAC: Simulation-Augmented Fine-Tuning of Open-Source LLMs for Analog Circuit Design | Advisor: *Huan Zhang*

- Led a project on post-training of open-source LLMs for specification-conditioned analog circuit design.
- Constructed a simulation-grounded dataset of 8,626 design tasks across 10 analog circuit types, covering textual specifications, labeled I/O ports, circuit loads, testbenches, target metrics, and reference netlists.
- Designed a three-stage SFT curriculum for circuit knowledge acquisition, masked component prediction, and simulation-feedback-guided correction.
- Implemented a GRPO-based reinforcement fine-tuning pipeline that uses topology checks, BO-based sizing, and SPICE simulation outcomes as reward signals.
- Fine-tuned Qwen3-8B and achieved 71.4% single-pass pass@1 and 78.2% two-pass pass@1, outperforming strong proprietary and open-source LLM baselines.

MAC-Tuning: LLM Multi-Compositional Problem Reasoning with Enhanced Knowledge Boundary Awareness
Advisor: *Yi.R Fung*

- Led a project on improving LLM hallucination detection and reasoning under multi-problem compositional settings.
- Proposed Multiple Answers and Confidence Stepwise Tuning (MAC-Tuning), a fine-tuning method that improves answer precision by an average of 25%.
- Built the data collection and training pipeline, including multi-problem prompt construction, model fine-tuning, and output evaluation.
- Designed evaluation metrics including accuracy, AP score, and MAP to assess model performance under both answer correctness and confidence calibration.
- Implemented automatic evaluation using LLM-as-a-judge and keyword-based extraction, and manually verified selected cases for reliability.

Random Augmentations Cheaply Break LLM Safety Alignment | Advisor: *Gagandeep Sigh*

- Contributed to a project studying how simple random prompt augmentations affect the robustness of LLM safety alignment.
- Implemented and evaluated multiple low-cost input augmentation strategies, including string-level and character-level perturbations.
- Helped design evaluation metrics and manually labeled model outputs to compare LLM-based safety classification with human evaluation.
- Conducted case studies showing that inexpensive random augmentations can substantially increase the success rate of bypassing safety-aligned LLMs.

Publication

- Jason Vega, **Junsheng Huang**, Gaokai Zhang, Hangoo Kang, Minjia Zhang, Gagandeep Singh. *Stochastic Monkeys at Play: Random Augmentations Cheaply Break LLM Safety Alignment*
- **Junsheng Huang**, Zhitao He, Sandeep Polisetty, May Fung. *MAC-Tuning: LLM Multi-Compositional Problem Reasoning with Enhanced Knowledge Boundary Awareness*. Admitted to EMNLP 2025 Main.
- **Junsheng Huang**, Yifan Sun, Zhouer Zhang, Ning Wei, Yening Liu, Michael Molter, Pavan Kumar Hanumolu, Elyse Rosenbaum, Bin Hu, Huan Zhang. *SAFTAC: Simulation-Augmented Fine-Tuning of Open-Source LLMs for Analog Circuit Design*. Manuscript under review.

Projects

391 OS System

- **Basic Functionalities**: Implemented an operating system supporting basic functionalities like scheduling, interrupts, system calls, exceptions, and file systems
- **Self-designed features**: ATA drivers to support writable file system, command history, changeable color and auto complete

Technical Skills

Technical Skills: Python, C/C++, CUDA, PyTorch, Hugging Face Transformers, PEFT/LoRA, DeepSpeed/FSDP, vLLM, verl
Spoken Languages: English (Proficient), Mandarin (Native), Cantonese (Native)

Honors

Honorable Mention of Mathematical Contest of Modeling (**May 2023**)

ZJU-UIUC Institute Dean's List in **Semaphore Year**

ZJU-UIUC Institute Third-Class Academic Excellence Award for **Semaphore Year** and **Junior Year**

UIUC Grainger Engineering Department Dean's List for two semesters in **Junior Year**

Teaching and Leadership

Teaching Assistant for CS446/ECE449 (Machine Learning)	Jan 2026 - May 2026
Teaching Assistant for MATH285 (Introduction to Differential Equations)	Jan 2025 - Jun 2025
Teaching Assistant for MATH241 (Calculus III)	Sep 2024 - Dec 2024
Course Assistant for ECE391 (Computer System Engineering)	Jan 2024 - May 2024
Student Representative for ECE Major in ZJUI	Sep 2021 - Jun 2022, Sep 2024 - Jun 2025